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A novel transfer learning-based model for diagnosing malaria from parasitized and uninfected red blood cell images

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ABSTRACT

Malaria represents a potentially fatal communicable illness triggered by the Plasmodium parasite. This disease is transmitted to humans through the bites of Anopheles mosquitoes that carry the infection. This disease has significant and devastating consequences on the health systems of fragile countries, particularly in sub-Saharan Africa. Malaria affects red blood cells by invading and replicating within them, destroying them, and releasing toxic byproducts into the bloodstream. The parasite's ability to stick and modify the surface of red blood cells can cause them to become sticky, obstructing blood flow in vital organs such as the brain and spleen. Therefore, efficient approaches for the early detection of malaria are critical to saving patients' lives. The main aim of this study is to develop an efficient model for early malaria diagnosis. We used malaria images based on parasitized and uninfected red blood cells for the study experiments. We applied neural network-based approaches such as Neural Search Architecture Network (NASNet) and compared its performance with machine learning techniques. Moreover, we proposed a novel NNR (NASNet Random forest) method for feature engineering. The proposed NNR approach first extracts spatial features from input malaria images, then class prediction probability features are extracted from these spatial features. The feature set obtained from the data extraction trains machine learning models. Our comprehensive experiments show that the support vector machine outperformed state-of-the-art models, achieving a high-performance score of 99% and having an inference time near 0.025 s. We validated the performance using k-fold cross-validation and optimized the hyperparameters through tuning. Our proposed research has improved the early diagnosis of malaria and can assist medical specialists in reducing the mortality rate.

1. Introduction

Malaria is a severe illness caused by the Plasmodium parasite. It is mainly spread through the bite of infected females of Anopheles mosquitoes [1]. The disease is caused by various species of Plasmodium, such as Plasmodium vivax and Plasmodium falciparum. These parasites invade and multiply inside the red blood cells of humans [2,3]. Malaria remains a major health concern worldwide, notably in tropical and subtropical countries, affecting millions annually and causing serious damage to their economies. Typical indications of malaria encompass elevated body temperature, shivering, aching head, queasiness, retching, muscular discomfort, weariness, and chest discomfort. If patients are left untreated, malaria can cause serious complications to them and even lead to mortality.

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The late diagnosis of malaria can pose critical risks, potentially leading to the death of the patient. In 2016, the World Health Organization (WHO) reported that there were 212 million cases of malaria worldwide. By 2019, this number had increased to 229 million [4]. The annual mortality rate from malaria is estimated to be 409 thousand [5,6]. Studies indicate that malaria primarily affects young children under the age of five [7,8]. However, traditional microscopic examinations for accurate diagnosis take longer. The accuracy of malaria diagnosis depends on factors such as test costs, the sensitivity of the technique, attention to detail, the time required for the test, and the expertise of the person conducting it [9]. Therefore, there is a need for an advanced automated artificial intelligence-based model that can early and accurately detect malaria from human red blood cells.

In recent years, artificial intelligence (AI) has significantly improved early diagnosis of various critical and chronic diseases in the medical domain [10–13], saving and improving patients' lives [14]. Since traditional methods of malaria diagnosis, such as microscopic examination, can be time-consuming and require skilled personnel, AI-based models offer the potential alternative for faster and more accurate detection, aiding in timely diagnosis and treatment [15]. AI models can analyze digital images of red blood cells and identify malaria parasites with high precision. To accomplish this, we can employ deep learning approaches, such as convolutional neural networks (CNN) [16], to classify blood cells and effectively filter out infected cells. This approach is particularly valuable in rural areas where microscopic tests for malaria detection may be unavailable. These learning methods can be turned using several optimization methods, such as in [17–20].

This study proposes a transfer learning-based mechanism for efficiently diagnosing malaria. Transfer learning has emerged as a promising approach for malaria diagnosis using red blood cell images, explicitly focusing on spatial features [1]. Transfer learning allows for extracting significant traits from images and capturing low-level and high-level representations by exploiting pre-trained deep-learning models [21]. This approach facilitates the transfer of learned knowledge to new datasets, reducing the need for extensive labeled data and computational resources [22]. By incorporating transfer learning into diagnosing malaria, we can leverage the spatial features of red blood cell images, thereby enhancing the accuracy and efficiency of the diagnostic process. Ultimately, this can aid in the early detection and treatment of this life-threatening disease.

The primary research contributions of the study for the early diagnosis of malaria are summarized followed as:

- A novel approach called NASNet Random forest (NNR) is proposed based on features extracted from fine-tuning of pre-trained models following a transfer learning approach. The NNR technique extracts spatial features from input malaria images, specifically based on parasitized and uninfected red blood cells. These spatial features are then used to extract class prediction probability features, which are further utilized in building the applied machine learning methods.
- We evaluate the performance impact of using the NNR method on various machine learning and neural network techniques. All applied approaches are hyperparameter-tuned, and performance assessment is confirmed using k-fold cross-validation. Computational complexity analysis was also conducted to assess the efficiency of the applied techniques. Our study's results outperformed the state-of-the-art approaches.

The remaining sections of this research are formulated as follows: Section 2 presents an analysis of the related literature. Section 3 outlines our proposed methodology. Section 4 presents the comparative results of applied AI techniques. Finally, the study findings are summarized in Section 5.

2. Related work

In recent years, researchers have explored the potential of using automated image analysis techniques to aid in malaria diagnosis. Various studies have employed machine learning algorithms and image processing techniques to distinguish between parasitized and uninfected red blood cells based on their morphological characteristics and staining patterns. This literature analysis underscores the importance of continuing research in this field to develop robust and accessible diagnostic tools that can help healthcare providers identify malaria infections promptly and accurately. The summary of the analyzed literature is provided in Table 1.

In [1], Vijayalakshmi et al. employed a transfer learning strategy for recognizing the illness caused by the Falciparum malaria parasite using a unique Deep Neural Network (DNN) model. The study utilized the Support Vector Machine (SVM) and geometry group network to identify the severe form of the malaria parasite. Their evaluation data set consisted of microscopic images of stained blood smears. The proposed VGG-19 SVM model outperformed the CNN model. However, the performance score obtained from the VGG-19 SVM project model was 93. 13%, which is considered relatively low.

In [23], the study presented a framework for identifying malaria caused by the Plasmodium falciparum parasite. The experiments used a dataset obtained from the Parasitology laboratory at Gadjah Mada University. To identify the malaria parasite, an advanced feature extraction strategy based on histogram-based texture features was applied to the image data. The acquired characteristics were subsequently inputted into an Artificial Neural Network (ANN) [24] model to undergo training and testing. The algorithm used for training the ANN was the backpropagation algorithm of a multilayer perception [25]. The experimental performance results demonstrated that the proposed ANN achieved a performance accuracy score of 87.8% with a computation time of 0.55 s. However, the performance scores of the study were relatively low, indicating the need for more accurate diagnostic techniques.

Suraksha et al. [31] focused on classifying blood smear images of malaria parasites. The dataset used in their study is the protozoal red blood cells obtained from the National Library of Medicine. An investigation involving the categorization of malaria images employs a Convolutional Neural Network (CNN) model, which relies on principles of deep learning methodologies. Additionally, a hybrid model [32] combining a pre-trained VGG-19 and CNN is utilized. The VGG-19-CNN model achieves satisfactory results compared to other models, with an accuracy of 96.02%. However, there is still a need for further performance enhancement.

In [26], the study presented a segmentation strategy designed to identify microscopic images of the malaria parasite. The dataset of the images was obtained from the Eijkman Institute for Molecular Biology in Indonesia. The dataset consists of thirty images of microscopic blood smears for patients with malaria. The empirical findings demonstrate the significance of the deep segmentation method in handling different image and histogram conditions, resulting in improved overall performance. The proposed approach achieved an overall metric score of 86%. However, when compared to other studies, the performance results of their approach were relatively poor.

Alassaf et al. [33] developed a unique strategy called IDTL-MPDC, and accurately detected the malarial parasites in blood smear images. The dataset used for the study has blood smear images. Their approach includes different sub-steps: preprocessing based on median filtering (MF), feature mining based on Res2Net, hyperparameter tuning based on the differential evolution (DE) algorithm, and ranking based on K-nearest neighbor (KNN) results. The DE algorithm achieved an excellent accuracy score of 95.86%, which can be further improved.

In [28], Masud et al. used mobile clinical systems and deep learning to identify lethal malaria in patients. The main focus of their study was to utilize a CNN structure to effectively and accurately diagnose malaria using a mobile application. The dataset used for to evaluate

Table 1
Related literature summary analysis.

Ref.	Year	Dataset	Proposed technique	Performance accuracy
[23]	2016	Data from Parasitology laboratory of Gadjah Mada University	ANN deep learning with Feature extraction	87.8%
[1]	2020	Microscopic images from stained blood	DNN + VGG-19 SVM	93.13%
[26]	2019	Images of microscopic organized at the Eijkman Institute for Molecular Biology, Indonesia	Deep learning Segmentation method	86%
[27]	2022	Open source blood smear images dataset	Deep learning IDTL-MPDC	95.86%
[28]	2020	Malaria dataset from NIH	Deep learning + cyclical-SGD	97.30%
[29]	2020	National library of medicine dataset	DL-Xception with mish	98.86%
[30]	2022	Malaria blood sample images of balanced class dataset	DL-DAGNN	94.79%
[31]	2023	National library of medicine dataset	VGG-19-CNN	96.02%

their approach is publicly available on Kaggle and was obtained from Chittagong Medical College Hospital in Bangladesh, consisting of two hundred patient records. The suggested model achieved an accuracy score of 97.30% through performance evaluation of a traditional CNN model using cyclical-SGD.

Sriporn et al. [29], utilized malaria disease images to identify and detect malaria based on a CNN model. The dataset used in the study was obtained from the National Library of Medicine and consisted of 151 infected patients and 50 non-infected individuals. The Xception model was employed with the activation algorithm Mish and optimizer Nadam to improve the accuracy of ranking malaria illness in thin blood smear images. The approach utilized a total of 7000 images, with 4500 contaminated images and 2500 non-contaminated images. The efficacy achieved by Xception with Mish was 98.86%.

Oyewola et al. [30], introduced a novel method of deep learning named as data augmentation convolutional neural network (DACNN), which focuses on analyzing blood smear images of Plasmodium parasites to aid in identifying malaria illness in humans. The dataset used in the study is an open-source dataset downloaded from Kaggle. A comparative analysis is conducted between the suggested model and various CNN-based models to evaluate its performance. The results of this research indicate that the suggested approach exhibited superior performance compared to alternative deep learning models, achieving an optimum accuracy score of 94.79%. However, it should be noted that this accuracy score is relatively low when compared to other studies.

2.1. Research gap

Through this literature analysis, we have determined the following major research gaps that need to be covered:

- Although classical deep and machine learning approaches have been used extensively in malaria diagnosis, it is crucial to adopt advanced learning approaches such as transfer learning techniques. These advanced methods show remarkable performance, where they efficiently reuse pre-existing models, alleviating the challenge of limited patient data, and enhancing the stability of the diagnosis system.
- The current performance scores for malaria detection can be significantly improved by proposing novel transfer learning-based feature engineering mechanisms.

3. Proposed methodology

In this section, we discuss our novel research methodology for the Malaria diagnosis step-by-step (see Fig. 1). The proposed methodology utilizes malaria images of parasitized and uninfected red blood cells. We introduce a novel approach called NNR for feature engineering from malaria image data. This feature engineering process creates a new feature set, which is divided into training and testing sets using an 80:20 split ratio. We have applied advanced machine learning and neural network approaches with hyper-parameter optimization for comparison. The applied approaches are trained and tested to evaluate their performance. The model that outperforms the others is then used for diagnosing malaria from red blood cell images.

3.1. Parasitized and uninfected red blood cells data

This study utilized a benchmark dataset [34] that includes images of parasitized and uninfected red blood cells related to malaria. The sample images corresponding to each class are illustrated in Fig. 2. The dataset has 27,601 malaria images, as depicted in Fig. 3. The dataset has been carefully balanced to facilitate experimental analysis.

3.2. Novel proposed approach

This study introduces a unique NNR approach aimed at enhancing the feature engineering process using a transfer learning-based approach, as shown in Fig. 4. The image data of parasitized and uninfected red blood cells related to malaria are input to a pre-trained NASNet model for extracting spatial features. The spatial features obtained are subsequently utilized as inputs for the random forest technique to derive class prediction probability features [35]. The resulting features are used to build deep machine-learning models for malaria diagnosis. As we discuss later in Section 4.3, the study results show that the employed method demonstrated excellent performance results through the utilization of the suggested transfer features.

Transfer learning-based feature engineering presents significant advantages over classical methods in diagnosing malaria from parasitized and uninfected red blood cell images. Traditional feature engineering approaches often rely on handcrafted features that may not capture all the relevant information in the images, leading to limited discriminatory power. In contrast, transfer learning leverages pre-trained deep neural networks, enabling them to learn generic features that are broadly applicable across different tasks. This empowers our proposed

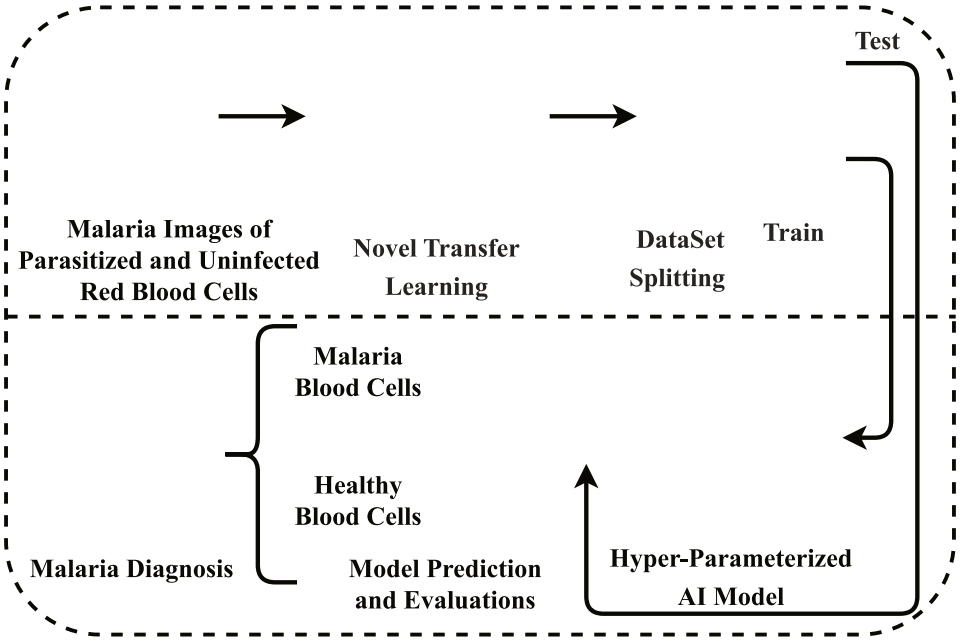

Fig. 1. Architecture of the proposed Malaria Diagnosis Method.

(a) (b)

Fig. 2. The red blood cell images data analysis with the target class.

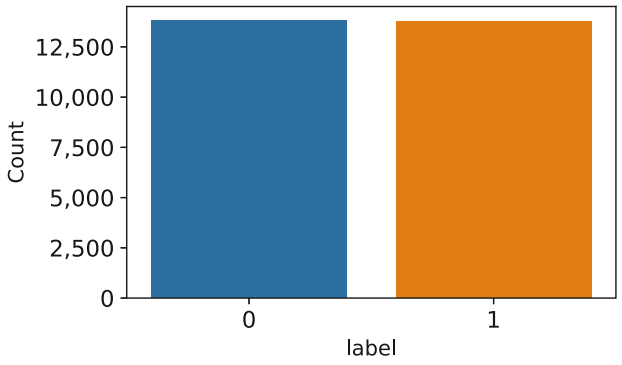

Fig. 3. Distribution of target labels based on the images of red blood cells in the benchmark dataset.

model to extract highly discriminative features from the red blood cell images, thereby enhancing its ability to distinguish between parasitized and uninfected cells. Overall, the implementation of our proposed transfer learning-based feature engineering represents a powerful approach to improving the accuracy and robustness of malaria diagnosis from red blood cell images.

Algorithm 1 shows the step-by-step flow of the proposed feature engineering approach.

Algorithm 1 NNR Method

Input: Malaria images.
Output: New transfer features for detecting malaria.
initiate;
1- F_{nasnet} $NASNet_{prediction}(MiS)$ // MiS Malaria images, here MiS is input image data and F_{nasnet} is newly obtained spatial feature set.
2- F_{rf} $RF_{probabilities\ prediction}(F_{nasnet})$ // here F_{rf} represents the collection of probabilistic-based features that are extracted.
3- F_{Prob} F_{rf} // hereF_{Prob} represents the conclusive set of probabilistic-based features employed for the identification of malaria.
end;

3.3. Machine learning techniques

Artificial intelligence techniques [36], such as deep and machine learning, have emerged as powerful tools for the diagnosis of malaria using images of red blood cells. These techniques leverage advanced algorithms to extract meaningful features from the images, enabling accurate classification of infected and uninfected cells. By training models on large datasets, artificial intelligence can learn complex patterns and variations associated with malaria, enabling early and precise detection. This automated approach not only reduces the